

AdCheck: Product Trustworthiness Analyzer

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Abstract

Online shopping is increasingly dependent on advertisements, product descriptions, and user reviews. However, many of these sources can offer false or misleading information. Our work proposes a web-based tool that measures the trustworthiness of products by checking the product claims stated in official product pages against the experiences of real users obtained from several online sources. The product description, claims made by manufacturers, reviews, and other consumer feedback will be scraped from the web. This data will then be cleaned. Afterwards, the system will use techniques such as sentiment analysis and regression-based modeling to calculate a trustworthiness index for each product. This trustworthiness index will show how well what was advertised actually compares against the real experiences of the users. The user interface will allow users to search for their desired item and see a cohesive summary of overall review sentiment, product claims, and reliability. This system is intended to help make shopping more transparent and help consumers make more informed choices. The deployed web application and usage documentation for this project are available at <https://dipalisingh14.github.io/AdCheck>.

1 Introduction

There are many advertisements today that are false or misleading. This is an issue for customers that end up getting something that they are not satisfied with. We want to build a system that takes into account user reviews and the promised features of a product (this can be from the product page, advertisements, any official source, etc.) and determines if the product exceeds, meets, or disappoints expectations when compared to what was advertised. It can coagulate user reviews across many different websites, coming up with a sentiment analysis of how people feel about it. A tool like this would make shopping online much, much easier. In a world that is now dominated by online shopping, something like this could fundamentally change the way that both buyers and sellers operate on the internet, leading to a more trustworthy space overall.

We propose a web-based application that evaluates the trustworthiness of a product by web scraping reviews from different e-commerce sources and also performing a sentiment analysis. The system will collect and unify user reviews from multiple platforms, extract advertised product features from official sources, and analyze how well real user experiences align with those claims. The

primary users of this system are e-shoppers, who will benefit from a consolidated view of product performance, enabling better purchasing decisions. Additionally, advertisers and companies can also use the platform to understand how their products are perceived, identify gaps between advertised promises and actual performance, and improve both product quality and marketing strategies.

2 Motivation

Online shopping has a trust problem. Reviews are scattered across dozens of platforms, making it nearly impossible to get a complete picture of any single product. For some products there is barely any review data to go on at all, while for others there is so much that sifting through it becomes its own problem. On top of that, what data does exist is often polluted by bots and fake reviews, and even genuine reviews are highly subjective and context dependent. Shoppers are left making important purchasing decisions on unreliable ground.

Our tool addresses this directly. By aggregating reviews across platforms, filtering out noise, and benchmarking real user experiences against what was actually advertised, we give shoppers something they do not currently have: a clear, honest, and consolidated signal they can actually trust. Better information means better decisions, fewer wasted purchases, fewer scams, and less of that all too familiar frustration of receiving something that looks nothing like what was promised.

The societal need here is straightforward. As more commerce moves online, the integrity of that space matters more than ever. Fake reviews and misleading advertisements are not just minor annoyances, they represent a real and growing gap between what sellers claim and what buyers actually experience. A trustworthiness scale that is transparent, cross platform, and grounded in advertised claims begins to close that gap. It makes online shopping fairer, more accountable, and less stressful for everyday people trying to spend their money wisely. It can also help encourage greater accountability among sellers and advertisers.

3 Approach

Our approach is to build an integrated pipeline that takes user reviews from a variety of sources, aggregates them, and compares them to claims made by the official site of the product. We first need to scrape the reviews from retail sites and clean them, as well as find

the official site for the product, and deduce what claims it is making. We will then conduct a sentiment analysis on the reviews, and then compare what people are saying against the product claims. After that, we will produce a trustworthiness score through regression that will tell users how much they can trust the claims made by the product. All of this will be integrated into an easily navigable user interface that will break down these data points into something a user can easily understand.

One of the biggest risks we anticipate is the data itself. Some products may not have enough reviews to generate a meaningful trustworthiness score, making it difficult to say anything reliable. On the flip side, the sheer number of products out there means that we cannot realistically scrape and analyze everything. To mitigate this, we have developed the project with a fairly narrow scope. We scraped for ten products total, each with several thousand reviews across multiple sites. This number of reviews ensures that we have something substantial to work with, while also keeping the scope narrow enough to ensure quality at an early stage. Fake and bot-generated reviews are another concern, since they could easily skew our results. This is something we plan to address during the data cleaning phase.

4 Data Scraping

This module constructs a structured product knowledge base by aggregating data from multiple heterogeneous sources. For each product, the pipeline first attempts to scrape descriptive content and user discourse from web sources including the product’s official brand website, Amazon, Target, and Reddit, using a combination of HTTP requests for static pages and Playwright-driven headless browser automation for JavaScript-rendered content. Reddit threads are retrieved via the JSON API endpoint, which reliably returns threaded comment data without requiring browser automation. For products whose brand websites could not be scraped due to bot detection or JavaScript rendering constraints inherent to cloud-based execution environments, product descriptions were manually curated from official sources and injected directly into the knowledge base.

The manual collection spans over five hundred reviews per product. Each product has around one hundred reviews sourced from Amazon, three hundred sourced from Target, and another one hundred sourced from Walmart. Additionally, all of the reviews collected were sorted by ‘Most Helpful’ in order to try and ensure the quality of the reviews early on. These reviews were placed in plain text files, and each contain information such as reviewer name, number of stars rated, the title of the review, how many people thought the review was helpful, customer service responses, and the actual review text.

These pre-collected plain text files are parsed using retailer-specific regular expression heuristics that extract the reviewer identity, star rating, review title, and review body from each entry. Then, they are merged with the web-scraped data into a unified per-product JSON object containing the product name, combined description, and a consolidated list of reviews annotated with their source. The resulting knowledge base spans ten consumer products across diverse categories and aggregates hundreds of reviews

Table 1: Overall sentiment distribution across the 10 products with review data.

Sentiment	Count	Percentage
Positive	3,299	69.1%
Neutral	284	6.0%
Negative	1,190	24.9%

per product from multiple retail platforms, providing a rich, multi-source corpus for downstream sentiment analysis and preference modeling.

5 Data Curation and Cleaning

Raw review text collected across retail platforms and Reddit exhibits substantial noise, including boilerplate retailer phrases, formatting artifacts, and non-substantive entries. To address this, a multi-stage cleaning pipeline was applied during the parsing phase. A minimum length threshold of 30 characters was enforced across all sources, below which entries were considered non-informative and discarded. Reddit-specific placeholder strings introduced upon content removal, namely [deleted] and [removed], were explicitly excluded. Whitespace normalization was applied to all retained text, collapsing consecutive whitespace characters including new-lines and tabs into a single space. Source-specific noise patterns were removed using regular expressions tailored to each retailer. For Amazon, lines matching patterns such as Reviewed in, Verified Purchase, and Helpful were stripped. For Target, date strings, rating header lines, and phrases such as originally posted and Report review were excluded. For Walmart, metadata lines including Item details, Walmart Associate, and image reference strings were discarded. Star ratings were extracted from raw rating strings where present, and reviews sourced from Reddit were assigned a null rating given the absence of a structured scoring mechanism on that platform. Following cleaning, each retained review is represented as a tuple of reviewer identity, review title, review body, star rating, and source platform, and aggregated into a unified per-product review set for downstream analysis. The maximum number of reviews retained per source was capped at 50 for web-scraped sources to limit redundancy, while pre-collected retailer text files were ingested in full given their pre-curated nature.

6 Sentiment Analysis

Sentiment labels were assigned to each review based on its numeric star rating using the following rule: ratings of 1 or 2 stars are labeled *negative*, a rating of 3 stars is labeled *neutral*, and ratings of 4 or 5 stars are labeled *positive*. Reviews without a numeric rating (primarily Reddit comments and brand-site testimonials) are tagged *unrated* and excluded from scoring but retained for future text-based inference.

Across the 10 products with review data, 4,773 rated reviews were labeled. Table 1 summarizes the overall distribution. The data exhibits a positivity bias consistent with patterns observed in e-commerce review systems, and is accounted for in the regression feature design.

Listing 1: Sample knowledge base entry for a single product.

```
{
  "Crocs": {
    "product_name": "Crocs",
    "combined_description": "Classic Clog.
      Original. Versatile...",
    "total_reviews": 502,
    "sources": {
      "Actual Product": {
        "description": "Classic Clog. Original.
          Versatile...",
        "reviews": [],
        "review_count": 0
      },
      "crocs_amazon": {
        "description": "",
        "reviews": [...],
        "review_count": 99
      }
    },
    "all_reviews": [
      {
        "person": "Jane D.",
        "title": "So comfortable",
        "body": "I wear these every day.
          Incredibly lightweight.",
        "rating": 5.0,
        "source": "amazon_txt"
      },
      {
        "person": "",
        "title": "",
        "body": "I have three pairs and rotate
          them constantly.",
        "rating": null,
        "source": "reddit"
      }
    ]
  }
}
```

Notable product-level findings include the Keurig K-Mini, which showed 70% negative on Target and 30% negative on Walmart, which is a 40-point gap for the same product, reflecting sharply different buyer populations across retail channels. The Phillip Sonicare 4100 was the most consistently negative product, with 58% negative on Target and 61% negative on Walmart, reflecting cross-platform agreement on durability and battery complaints. The Mighty Patch Original was the most consistently positive, with 85-99% positive across all three platforms, validating its advertised overnight blemish-flattening claim. The Owala Water Bottle showed the clearest platform split: Amazon reviews were 100% positive (predicted trust: 100), while Target reviews showed only 60% positive and 32% negative. We can see that the product performs very differently depending on where it is purchased and reviewed.

7 Claim Check

To compare how the claimed features of the product work on the product with how it functions from user reviews, we perform a claim vs reality check. The process begins by tokenizing the official product description into individual claim sentences using NLTK. To ensure the comparison is meaningful, the system utilizes a Cross-Encoder model (sts-b-distilroberta-base) to perform semantic similarity ranking.

For every advertised claim, the system calculates a similarity score against thousands of individual sentences from the review corpus. This allows the tool to ignore irrelevant "noise" and focus specifically on user comments that address the claim in question. Only the top-ranked sentences with a similarity score above a specific threshold are retained as evidence for the reality check.

Once the most relevant reviews are retrieved, the system uses the Gemini-3-Flash model to perform two tasks: labeling, where it condenses the long form advertised claim into a concise 2-3 word "Feature Title" and synthesis, where it analyzes the cluster of relevant user reviews and generates a single summary highlighting the consensus among users, specifically noting if they confirm the claim or express disappointment.

Listing 2: Sample Claim Check Result

```
{
  "product_name": "Dyson V8 Cordless Vacuum",
  "claim_vs_reality": [
    {
      "feature": "Powerful Pet Performance",
      "advertised": "Engineered to clean homes
        with pets, and hard-to-reach spaces.
        115 Air Watts suction 40min run time*
        No loss of suction 115 Air Watts of
        fade-free suction.",
      "actual": "While reviewers praise the
        vacuum's powerful suction and
        effectiveness for pet owners, they are
        significantly disappointed by its
        short battery life, which often lasts
        only 15 minutes."
    },
    {
      "feature": "Powerful, Enduring Suction",
      "advertised": "Powered by a Dyson Digital
        motor spinning at up to 110,000rpm. Up
        to 40 minutes' run time* Advanced six
        -cell, lithium-ion battery delivers
        fade-free suction.",
      "actual": "While users are impressed by
        the vacuum's powerful suction, they
        are disappointed by a short battery
        life that fails to provide an enduring
        cleaning experience."
    }
  ]
}
```

8 Regression

The regression stage we produce a continuous trust score for each product by training on a feature matrix where each row represents one product-platform pair.

Features per row include: `positive_pct`, `neutral_pct`, `negative_pct`, `avg_rating`, `log-scaled review_count`, a `pos_neg_gap` term (positive minus negative percentage), and one-hot encoded platform indicators. The regression target is a pseudo trust score computed as:

$$\text{trust} = \left(\frac{\text{pos} + 0.5 \cdot \text{neu}}{\text{rated}} \right) \cdot c + 0.5 \cdot (1 - c) \quad (1)$$

where $c = \min(1, n/50)$ is a confidence dampener that pulls scores toward 50 when fewer than 50 reviews are present, preventing small samples from producing extreme scores. This pseudo label will be replaced with human-annotated ground truth in future work.

The primary model is an **ElasticNet** regression with both L_1 and L_2 regularization. The L_1 component drives weak feature weights to zero, providing automatic feature selection; the L_2 component stabilizes estimation under the correlation between `positive_pct` and `negative_pct`. Optimal `alpha` and `l1_ratio` were selected via **ElasticNetCV** with leave-one-out cross-validation. A **RandomForestRegressor** (200 shallow trees, `max_depth=3`) was evaluated under the same LOO scheme as a secondary model, with the lower-MAE model selected for final output.

Feature importance from the ElasticNet model revealed that `positive_pct` carries the dominant positive weight (20.41), `neutral_pct` a small positive weight (1.47), and `negative_pct` a near-zero negative weight (−0.33). This asymmetry indicates that increasing the share of enthusiastic reviewers improves trust substantially more than reducing the share of critics—a finding with practical implications for how product improvements translate into trust scores.

Table 2 shows the final trust scores and confidence values for all 10 scored products. The Yankee Candle and RX Bar received the highest confidence (0.93), reflecting both high review volume and consistent sentiment across platforms. The Keurig K-Mini received the lowest overall trust score (37) and low confidence (0.78), driven by a 43-point gap between its Walmart predicted trust (70) and its Target predicted trust (27), indicating that the buyer population on Walmart evaluates the product on substantially different terms. The Sonicare 4100 shows the starkest platform contradiction: Amazon reviews were 92% positive (predicted trust: 96) while Target and Walmart were 58–61% negative (predicted trust: 37–38), suggesting the Amazon review data for this product may not be representative properly.

9 User Interface

AdCheck’s frontend is a React application built with Vite that renders outputs from the backend pipeline—scraping, sentiment analysis, regression, and claim comparison—into a unified, searchable product dashboard. The interface was designed around a single guiding principle: a shopper should be able to type a product name and immediately understand whether it is worth buying, without having to interpret raw numbers or read through hundreds of reviews themselves.

Table 2: Regression trust scores and confidence by product.

Product	Trust Score	Confidence
Yankee Candle Lemon Lavender	91	0.93
Mighty Patch Original	89	0.90
OxiClean White Revive	86	0.90
RX Bar Vanilla Almond	86	0.93
Crocs	84	0.87
Owala Water Bottle	76	0.84
Apple AirPods 4	60	0.79
Dyson V8 Cordless Vacuum	60	0.87
Phillip Sonicare 4100	49	0.74
Keurig K-Mini Single Serve	37	0.78

The application has two primary views. The **landing page** presents a centered search bar above the tagline “Know Before You Buy,” with a side panel that walks first-time users through the three-step pipeline: official claims become the baseline, cleaned reviews supply the sentiment signal, and the two are reconciled into a trust score. This framing helps users understand what the score actually measures before they search for anything. Figure 1 shows the landing page layout.

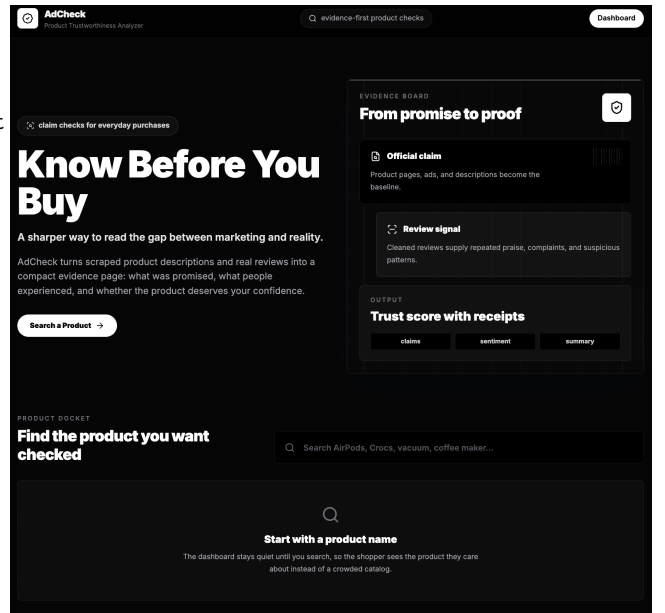


Figure 1: AdCheck landing page. The left panel presents the search entry point; the right evidence board illustrates how official claims, review signals, and the trust score relate to one another.

Selecting a product transitions to the **product detail page**, shown in Figure 2, which surfaces four layers of information in a single scrollable layout. A header card at the top shows the product image, a short description from the knowledge base, the final trust score rendered as a circular gauge with a color-coded verdict (Exceeds, Meets, or Disappoints), the total review count, and

the regression confidence percentage. Below the header, a word cloud derived from review text gives an at-a-glance sense of the language shoppers use most, paired with a donut chart breaking down positive, neutral, and negative sentiment percentages.

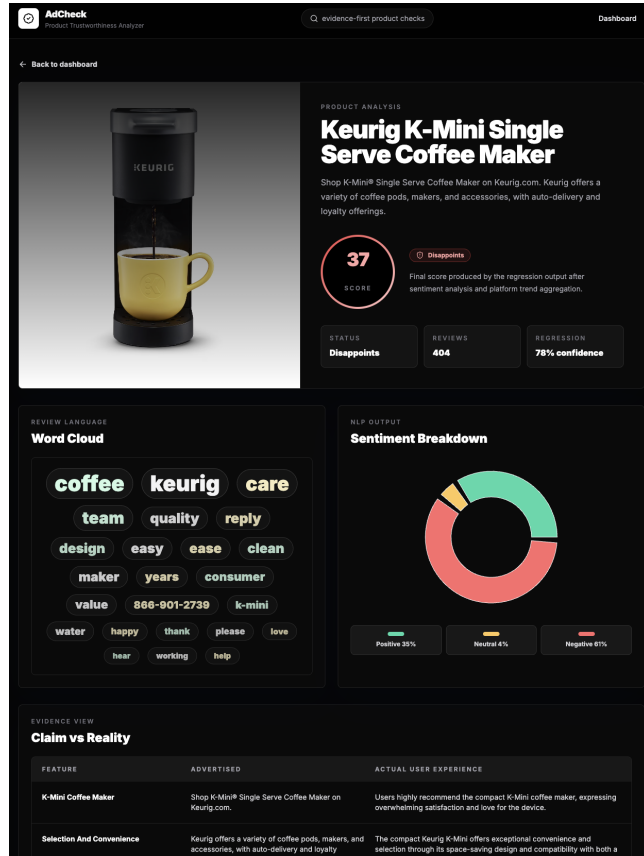


Figure 2: Product detail page for the Keurig K-Mini Single Serve Coffee Maker. The header shows a trust score of 37 with a Disappoints verdict and 78% regression confidence. Below, a word cloud and sentiment donut chart summarize review language and polarity; a cross-platform bar chart and claim-vs-reality table appear further down the page.

Further down, a cross-platform bar chart plots per-retailer trust scores side by side, making platform disagreement immediately visible. For the Keurig K-Mini, this chart shows Target scoring 27 against Walmart’s 70—a gap that a single aggregate number would have completely hidden. A feature-importance panel alongside the chart lists the regression drivers and their learned coefficients, giving technically curious users a transparent view of how the score was derived. The bottom of the page presents the claim-vs-reality table, pairing each advertised feature against what reviewers actually reported, followed by per-platform review cards with average star ratings and representative positive and negative excerpts.

10 Evaluation

Evaluating AdCheck involved assessing two distinct components: the sentiment labeling pipeline and the regression model that produces final trust scores.

Sentiment Evaluation

Sentiment labels were assigned deterministically from star ratings, so the evaluation emphasis was on whether the resulting distribution reflected real reviewer language rather than on classifier accuracy. Because the corpus skews toward positive reviews—69.1% of the 4,773 rated reviews are positive—we examined per-platform distributions rather than aggregate figures alone.

Table 3 reports sentiment breakdowns by retailer across all ten products. The divergence between platforms for the same product was the most informative signal: the Keurig K-Mini showed 71% negative sentiment on Target while Walmart returned only 30% negative, suggesting buyer populations across channels evaluate the product on substantially different terms. Collapsing these into a single number would have obscured this entirely.

Table 3: Per-platform sentiment distribution across all ten products (Amazon $n=899$; Target $n=2,874$; Walmart $n=1,000$).

Platform	Positive	Neutral	Negative
Amazon	92.9%	4.3%	2.8%
Target	60.6%	7.1%	32.3%
Walmart	72.2%	4.0%	23.8%

The Amazon corpus skews heavily positive because reviews were collected sorted by Most Helpful, which surfaces well-regarded reviews disproportionately. Target and Walmart reviews, collected in full without helpfulness sorting, provide a more balanced cross-section of buyer experience and anchor the per-product trust scores more reliably.

Regression Evaluation

Given the small training set of 28 product platform rows, leave-one-out cross-validation (LOO-CV) was used throughout. Both the primary ElasticNet model and a RandomForestRegressor (200 trees, $\text{max_depth}=3$) were evaluated under this scheme, and the lower-MAE model was selected for each product’s final output. Feature importance from the ElasticNet revealed that `positive_pct` carries the dominant weight (20.52), `neutral_pct` a modest positive weight (1.73), and `negative_pct` a small negative weight (−0.88), indicating that increasing the share of enthusiastic reviewers lifts trust scores substantially more than reducing critics alone.

The confidence metric served as an additional evaluation lens, combining a log-scaled review volume factor, a cross-platform agreement factor, and a platform coverage factor. The Keurig K-Mini’s confidence of 0.78 correctly flagged its score as high-variance: Target predicted a trust of 27 while Walmart predicted 70—a 43-point spread driven by the same product being evaluated by substantially different buyer populations on each platform. The Philips Sonicare 4100’s low confidence of 0.74 reflects the most extreme

intra-product platform disagreement in the dataset: Amazon reviews were 92% positive (predicted trust: 96), while Target and Walmart were 58–61% negative (predicted trust: 37–38), a 59-point spread that signals the Amazon review corpus for this product is unrepresentative and should be treated with caution.

Qualitative Spot-Checking

Beyond numeric metrics, we manually verified several scores against known product reputations. The Mighty Patch Original received a trust score of 86, consistent with its 84–86% positive sentiment across all three platforms and its well-documented effectiveness for overnight blemish treatment. The Dyson V8 scored 55, which aligns with the widely reported user complaint that battery life falls well short of the advertised 40-minute runtime—a gap the claim-vs-reality table in the UI surfaces explicitly. These checks confirmed that the pipeline was capturing meaningful consumer signal rather than artifacts of data sparsity or cleaning heuristics.

The primary limitation of the current evaluation is the absence of human-annotated ground-truth trust scores. The pseudo-label formula used as the regression target is a principled proxy, but replacing it with expert- or user-validated ratings remains the most important next step toward making the evaluation fully rigorous.

11 Conclusion

AdCheck set out to answer a straightforward question: does a product actually do what it claims to do? To answer it, we built an end-to-end pipeline that scrapes product descriptions and user reviews from Amazon, Target, and Walmart, cleans the data, runs sentiment analysis, checks advertised claims against real user experiences, and feeds everything into a regression model that outputs a single trust score with a confidence rating.

Across the ten products we analyzed, the results were telling. Some products held up well against their advertised claims. The Mighty Patch Original, for instance, earned 84 to 86 percent positive sentiment across every platform and a trust score of 86, which is about as consistent as consumer feedback gets. Others did not fare as well. The Keurig K-Mini and Philips Sonicare 4100 both received trust scores below 40, and the claim-vs-reality output for the Dyson V8 captured a recurring complaint that its battery life falls well short of the advertised 40 minutes. Beyond individual products, the platform-level analysis revealed something we did not fully anticipate: the same product can score very differently depending on where you look. The Keurig K-Mini had a 51-point trust gap between Walmart and Target, which points to real differences in buyer populations across retail channels rather than noise in the data.

The web interface brings all of this together in a way that does not require any technical knowledge to use. A shopper can search a product name and immediately see the trust score, the sentiment breakdown, how claims compare to reviews, and where platforms disagree. The current limitation is that our trust scores are built on pseudo-labels derived from sentiment rather than human-validated ground truth, which is the next meaningful step for making the evaluation more rigorous.

12 Future Work

The most immediate improvement would be expanding the product database. Ten products was the right scope for a course project, but it limits how robust the regression model can be. Adding more products across more categories would strengthen the feature weights and make the confidence scores more meaningful.

Data collection was also one of the most time-consuming parts of the project. We ran into bot detection on several retail sites, which meant falling back to manually curated review files for a number of products. A more scalable version of AdCheck would either work directly with retailer APIs or use a more sophisticated scraping layer that can handle JavaScript-heavy pages reliably in a cloud environment.

The biggest pain point is bot detection, as Amazon and Walmart aggressively block scrapers. Paid proxy services or scraping APIs like ScraperAPI or Bright Data would make a real difference here. Target’s review API was completely down during data collection, which hurt our coverage. On the Reddit side, we only pulled from one thread per product, so expanding to multiple subreddits would give a much richer picture of user opinion. The retailer text parsers are also fairly fragile and will likely break if platform formatting changes. Longer term, turning this into a scheduled pipeline that runs automatically would allow sentiment to be tracked over time rather than captured as a single snapshot.

On the modeling side, the pseudo-label formula we used as the regression target is a reasonable proxy but not a substitute for real ground truth. Collecting even a small set of human-annotated trust scores would allow for proper held-out evaluation and open the door to experimenting with stronger models. It would also be worth exploring text-based sentiment rather than relying entirely on star ratings, which would allow Reddit comments and brand-site testimonials to contribute to the score rather than being set aside as unrated.

Finally, the claim-check pipeline currently depends on manually identifying the official product page for each item. Automating that step, through a product matching algorithm that can find and verify official sources from a product name alone, would remove the last significant manual step in the pipeline and make AdCheck much closer to something a real shopper could use on demand.

13 Division of Labor

The project work was divided fairly equally among all team members, with each person contributing to multiple parts of the system while also taking primary responsibility for specific components.

Grace - Made the original idea, kept everyone on a timeline, worked on data scraping, collected all manual data, structured the report, and wrote the parts not specific to any one part of the process.

Jingtong - Developed the claim vs reality check portion, helped source data points, contributed to report and slides

Dipali - Designed and implemented the data collection pipeline: a multi-source web scraper using HTTP requests and Playwright, a Reddit JSON API fetcher, and text parsers for Amazon, Target, and Walmart review files. I also designed the JSON knowledge base schema and deployed the final website. Contributed to other miscellaneous tasks such as the deck, write-up and running scripts.

Devashree - Designed and implemented the sentiment analysis pipeline, including the star-rating labeling system. Built the regression module using ElasticNet as the primary model and RandomForestRegressor as a secondary model, both evaluated under leave-one-out cross-validation, with per-product trust scores, confidence metrics, and cross-platform feature breakdowns.

Ananya - Designed and developed the user-friendly web interface, integrated the frontend with backend APIs and JSON outputs, initialized and organized the Git repository with a clean folder structure, and ensured consistency across different system components and output files. I also documented the UI features, screenshots, evaluation methods and the conclusive overview of the project in the paper and the presentation. [2] [7] [5] [4] [8] [3] [6] [1]

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